XINYUAN ZHANG

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EDUCATION

Ph.D. in Analytics, Information Technology, Analytics, and Operations Department, University of Notre Dame
Aug 2022 – Present

- Cumulative GPA: 4.0/4.0
- Advisor: Ahmed Abbasi, Jeff Cai
- Research topics: Large-scale user digital trace data analysis with statistics, machine learning, and deep learning

M.S. in Statistics, University of California, Los Angeles Oct 2020 – June 2022

- Cumulative GPA: 3.93/4.0
- Supervisor: Mark S. Handcock
- Thesis: *An Awareness Model for a Two-sided Matching Market*

Bachelor of Commerce (Major in Finance and Statistics), **The University of Sydney** Feb 2017 – Dec 2019

- Cumulative GPA: 3.88/4.0
- Dean's Undergraduate Scholarship (International) 2017-2019

PUBLICATION AND WORK IN PROGRESS

Journal Publications

1. Bevilacqua, M., Oketch, K., Qin, R., Stamey, W., **Zhang, X.,** Gan, Y., Yang, K. Abbasi, A. "When Automated Assessment Meets Automated Content Generation: Examining Text Quality in the Era of GPTs," *ACM Transactions on Information Systems*, 43(2), 2025, no. 2, pp. 1-36. The all-time second most downloaded ACM TOIS paper (over 30K downloads).

Journal Manuscripts Under Review

1. **Zhang X**., Cai J., Li J., Abbasi A., "Designing for Digital Traces: Towards Holistically Contextualized Tensor-based Representations" Major Revision at *INFORMS Information Systems Research*.
2. Alavo R, **Zhang X**., Ademaj G., Cai J., Kwon H., Cotes R., Clifford G., Abbasi A., "Dash-M5H: An Interactive Dashboard for Multi-Modal, Multi-Model Mental Health Assessment", Submitted to *ACL 2026 demo*

Journal Manuscripts in Progress

1. Ademaj G., **Zhang X.,** Abbasi A., Sarker S., Sarker S., "Sensemaking in Multimodal, Multi-model Environments: Designing Support for Remote Mental Health Assessments." to be submitted to *INFORMS Management Scienc*e by August 2025.
2. **Zhang X.**, Cai J., Abbasi A., "Theory guided Multi-Source In-Context Learning Framework for LLM-based Health Assessments"
3. **Zhang X.,** Yoo E., Pedraza-Martinez A., "Volunteer Behavior in Online Communities."

CONFERENCE PRESENTATIONS

Ademaj G., **Zhang X.,** Abbasi A., Sarker S., Sarker S., "Sensemaking in Multimodal, Multi-model Environments: Designing Support for Remote Mental Health Assessments." presented at *Workshop on Information Technologies and Systems 2024*, *ICIS 2025*

Zhang X., Yoo E., Pedraza-Martinez A., "Volunteer Behavior in Online Communities." presented at *Productions and Operations Management conference 2024.*

Zhang X., Cai J., Li J., Abbasi A., "Towards a Holistic Representation of Online Customer Journeys: A Tensor-based Framework." presented at *INFORMS Workshop on Data Science 2023, INFORMS Workshop on Data Science 2024, ICIS 2025*

SELECTED RESEARCH EXPERIENCE

Research Assistant
Human-centered Analytics Lab (HAL), University of Notre Dame Aug 2022 – Present

- **Tensor-based Analysis of Digital Trace Data** (joint project with UVA)
  - Develop a novel 3D tensor-based framework for modeling digital user-channel interactions across time and space
  - Analyze 24 million raw online customers digital trace data spanning a 2-year period
  - Develop statistical and machine learning models to predict customer purchase decisions with parsimonious user journey representations
  - Analyze millisecond level multi-modal data from remote mental health assessment
  - Develop statistical and machine learning models to predict patient mental health outcomes
- **Examination and Comparison of Text Content Generated by Humans versus GPTs**

- Designed 50 prompts and generated 500 GPT-based essays as benchmark datasets
- Analyzed the effect of prompt type, prompt respondent, and ML model used on the ratings of GPT-generated contents with statistical models

Research Assistant
Department of Statistics, University of California, Los Angeles July 2021 – June 2022

- **Revealed Preference Model with Continuous Covariates**
  - Assisted with extending the revealed preference model (RPM) under the continuous covariates setting
  - Contributed to the update the rpm R package for RPM with both categorical and continuous covariates
  - Implemented various simulation studies, tested and verified the validity of the current updated model

Research Assistant
Behavioral Finance Lab, Tianjin University Dec 2018 – Feb 2019

- **Big Data Based Financial Innovation and Risk Analysis**
  - Implemented the Fama-French Model to study how individual investors' sentiments would affect the asset pricing process and transaction behaviors in the stock market

Teaching Experience

Course Instructor:
ITAO 30250 Data Wrangling with R Spring 20256 University of Notre Dame
ITAO 20200 Foundation of Statistics Spring 2025, University of Notre Dame

Professional Experience

Reviewed Manuscripts for the Following Journals and Conferences:
Journals: Manufacturing & Service Operations Management.
Conference: Pacific Asia Conference on Information Systems (2023); International Conference on Information Systems (2024, 2025); Hawaii International Conference on System Sciences (2024); INFORMS Data Science Workshop (2024, 2025), DESRIST (2026)

SKILLS

Computer Skills: Python (TensorFlow, sklearn, PyTorch), R, STATA, Latex